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A Comprehensive Introduction to 5G Network Architecture

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This article introduces a massive topic: 5G network architecture. To cover the entire 5G network in detail would require three to four hundred articles, so this post will provide a high-level overview of the entire system, focusing particularly on common terminology. We will fill in all the specific details in subsequent articles. Let's get started!

The Three Core Components

The 5G network architecture primarily consists of three core parts: **User Equipment (UE)**, **Radio Access Network (RAN)**, and **Core Network (CN)**. Let's take a look at what each of these entails:

- **User Equipment (UE):** These are the devices used to connect to the 5G network. For example, your smartphone, but nowadays it contains much more, such as various Internet of Things (IoT) devices and industrial sensors inside factories.
- **Radio Access Network (RAN):** This is the bridge connecting the UE and the CN. Simply put, it's the base station. In the 5G era, base stations are called **gNB** (Next Generation Node B, pronounced "g-node-b"). When a UE transmits a signal, it is received by the gNB; similarly, signals transmitted by the gNB are received by the UE. While these two actions sound identical, the underlying mechanisms are quite different. To distinguish them, UE transmitting to gNB is called **Uplink**, while gNB transmitting to UE is called **Downlink**. Roughly speaking, you can think of it as the difference between uploading and downloading. The differences go beyond just reversing the transmission direction because a single gNB typically connects to multiple UEs (meaning many phones connect to one base station). From the UE's perspective, it always has a single target gNB for transmission. But for the gNB, it must manage multiple connected UEs simultaneously. Additionally, the gNB is split into a **Central Unit (CU)** and a **Distributed Unit (DU)**. We will explain how and why this split happens later in the article.
- **Core Network (CN):** This is the brain of the 5G network. It handles essential tasks such as allocating IP addresses to connected UEs and authenticating whether your UE is authorized to connect. Broken down further, the core network includes the **UPF** (User Plane Function), **AMF** (Access and Mobility Management Function), **SMF** (Session Management Function), and more. Each part has its own specific duties, which we will cover later.

Standalone (SA) vs. Non-Standalone (NSA)

When discussing 5G, you will frequently hear two terms: Standalone (SA) and Non-Standalone (NSA). They exist because transitioning from a 4G to a 5G network means you can't simply replace the entire existing 4G infrastructure with 5G overnight. This led to the creation of a "hybrid combination."

This hybrid essentially mixes and matches 4G and 5G base stations and core networks, resulting in various combinations. However, when we refer to NSA, we usually mean replacing the base stations with 5G gNBs while keeping the 4G Core Network. SA, on the other hand, means both the base stations and the core network are purely 5G. Intuitively, this means NSA cannot unleash the full potential of 5G because its underlying brain (CN) is still 4G, though upgrading to gNBs does allow for some capabilities 4G didn't have. The ideal scenario is, of course, SA, where the fully 5G infrastructure can achieve maximum network flexibility.

Three Major 5G Application Scenarios

Who is 5G serving, and what are its goals? In short, 5G targets three primary use cases:

- **Enhanced Mobile Broadband (eMBB):** This is the most relevant to our daily lives. Simply put, it means "faster internet." Applications like streaming high-definition video or using VR/AR devices require massive bandwidth to transmit data.
- **Ultra-Reliable and Low Latency Communications (URLLC):** This focuses on extreme stability and real-time responsiveness. It's used in critical scenarios like remote surgery or autonomous driving, where a few seconds of network lag could lead to disaster.

- **Massive Machine Type Communications (mMTC):** Designed for smart city scenarios, the network must support a massive number of devices connecting simultaneously. A small geographic area might contain thousands of connected sensors — thermometers, streetlights, smart water meters, etc.

You might be thinking that achieving all three of these conditions simultaneously sounds impossible. You're right! That's why a technology called **Network Slicing** was introduced.

Network slicing essentially carves out multiple independent virtual logical networks on a single set of hardware infrastructure. It's like drawing dedicated lanes on a highway: one lane for eMBB users watching 4K video, one for URLLC autonomous driving, and one for mMTC smart cities. These lanes don't interfere with each other, effectively compartmentalizing network resources.

To let the system know which tier of service each application requires, telecommunications introduced the concept of **QoS (Quality of Service)**. You can think of QoS as a priority shipping label on a package. For instance, 4K video requires "high bandwidth," while autonomous driving demands "ultra-low latency." Every transmitted packet carries a QoS label, allowing the system to route it into the correct lane.

This slicing is **End-to-End**, meaning the dedicated lanes are established all the way from the CN, through the gNB, right down to the UE, ensuring strict hardware isolation. Imagine it's New Year's Eve, and tens of thousands of people are uploading videos, causing a massive traffic jam on the eMBB slice. Because of network slicing, this congestion will absolutely not spill over into the adjacent URLLC slice. This ensures that even during peak network traffic, critical commands for remote surgeries or self-driving cars remain perfectly unobstructed.

Why Split the Base Station into CU and DU?

Earlier, we mentioned splitting the base station into a CU and DU. Let's explain why. In the 4G era, the base station was called an **eNB** (Evolved Node B). It operated as a "black box" where all functions were bundled together. However, 5G demands much higher communication quality, so 3GPP (the organization responsible for telecom standardization) logically divided the gNB into two parts:

- **Central Unit (CU):** Responsible for "time-insensitive, non-real-time" high-level communication protocols. For example, when you turn your phone on to connect, the base station needs to verify your identity or encrypt your data. If these actions are delayed by a fraction of a second, you won't notice any lag. Because these tasks are time-insensitive, the CU can be placed in a remote server room far from the antennas, connected via cables. This allows for centralized management and lets multiple base stations share a single CU, significantly reducing deployment costs.
- **Distributed Unit (DU):** Responsible for "time-sensitive, real-time" low-level protocols. These protocols must react to signals instantly, so the DU must be placed very close to the antenna (**RU, Radio Unit**). For example, if electromagnetic interference causes your phone to miss a fragment of a signal, the network must detect it and request a retransmission within a millisecond; otherwise, your video or voice call will stutter. This "split-second rescue" work must be handled by the DU closest to the antenna.

Interfaces and Transmission Media

How do these separated components (RU, DU, CU, and CN) communicate? In reality, most wireless network infrastructure is actually wired; only the link between the UE and gNB uses electromagnetic waves over the air, known as the **Air Interface (Uu)**.

When the gNB's RU (antenna) receives a signal, it passes it to the DU via the **Fronthaul**. The DU processes it and sends it to the CU via the **Midhaul**. The CU then forwards it to the CN via the **Backhaul**. The fronthaul, midhaul, and backhaul almost exclusively rely on fiber optic cables; high-frequency wireless microwave transmission is only considered in rare remote areas where laying fiber is impossible. Inside the Core Network data centers, the AMF and SMF communicate using high-speed Ethernet or fiber patch cords.

Another major benefit of this architecture is preventing vendor lock-in. Previously, base stations ran on proprietary closed architectures, forcing telecom operators to buy everything from a single vendor with no room for mixing and matching. By splitting the CU and DU and standardizing the interface between them, 5G gives operators the freedom to "mix and match." They can buy CU software from Vendor A and DU hardware from Vendor B, spurring market competition, increasing bargaining power, and driving down network construction costs.

Control Plane (CP) vs. User Plane (UP)

The concepts of high-level and low-level protocols might still seem abstract. Let's introduce them properly so you can clearly visualize the CU and DU split.

In 5G, you don't just magically turn data into radio waves and blast it out of an antenna. There are countless complex issues to solve. What if the UE sends a signal but the gNB misses it? How does the UE know to resend it? If a user is on a high-speed train moving from Base Station A to Base Station B, how do the stations coordinate a handover? How does a phone initially attach to a network when turned on?

The most intuitive way to manage this chaos is to separate these tasks into different functional layers. By defining exactly what each layer is responsible for, the system stays organized. A foundational concept in telecommunications is the separation of the **Control Plane (CP)** and **User Plane (UP)**. While all signals leave the UE's antenna, they travel different routes conceptually:

- **Control Plane (CP):** The "command and management" system. It transmits **Signaling** — control data responsible for tasks like network attachment and handover management.
- **User Plane (UP):** Handles the actual user data we care about, like watching YouTube or sending text messages.

Next, let's take a look at their internal protocol layers. Some protocol layers are exclusively traversed by the control plane, while others are traversed by both the control plane and the user plane. "Traverse" here means that once a signal is generated, it sequentially passes through all these protocol layers, completing the corresponding tasks at each layer before moving on to the next, much like the OSI model. Figure 1 is the official 5G architecture diagram provided by 3GPP.

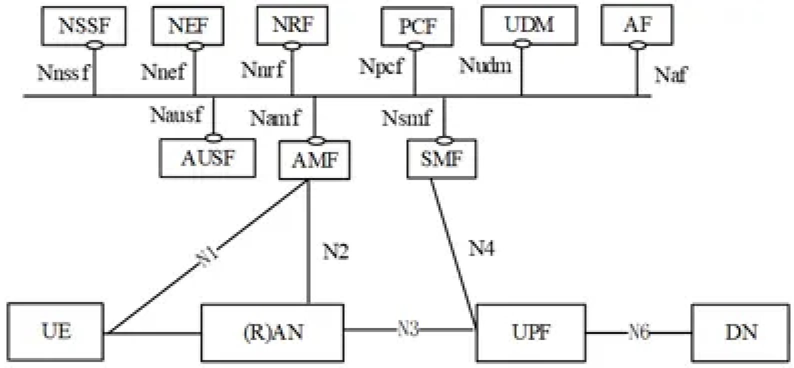

Figure (1): 5G System Architecture (https://www.3gpp.org/technologies/5g-system-overview)

Does it look incredibly complex to you? Let's break it down one by one. UE and RAN have already been introduced, but what about the Core Network (CN)? In Figure 1, everything other than UE, RAN, and DN is actually part of the CN. This diagram precisely illustrates the concept of separating CP and UP. The user plane follows the path at the very bottom of the diagram, namely the UE-RAN-UPF-DN route.

Let's briefly explain the DN (Data Network) on the far right of the diagram. The DN is the ultimate destination or source of our data. For ease of understanding, we usually think of it as the "Internet" we use daily — for example, when you watch a YouTube video, the video packets are sent from the Internet (acting as the DN) to the UPF. Then, the UPF transmits them to the RAN through an interface called N3, and the RAN forwards them to your UE.

Next, let's look at the upper half, which is where commands are issued. The core units in the diagram are the AMF (Access and Mobility Management Function) and the SMF (Session Management Function). The AMF communicates with the UE through an interface called N1 and with the base station through N2. Whether a phone has permission to access the internet and which base station's coverage area it moves into are all managed by the AMF. The SMF acts as a connection dispatcher; when you are ready to go online, the SMF allocates an IP address and tells the UPF via the N4 interface that user data is about to be transmitted.

The N1~N6 elements in the diagram merely represent connection formats. They essentially establish what needs to be transmitted and the data format

challenge lies in the "radio wave" transmission between the phone (UE) on the far left and the base station (RAN). To ensure signals are delivered smoothly and securely, both the phone and the base station have a corresponding set of built-in "protocol layer" processing pipelines.

When a phone wants to send data, the data is processed and "packaged" layer by layer from top to bottom inside the phone. When the base station's antenna receives the electromagnetic waves, it "unpacks" and interprets them layer by layer from bottom to top. The two sides communicate through these layers, which correspond one-to-one, with each performing its specific duties.

In the telecommunications field, this "packaging" and "transmitting" process involves three terms:

- **SDU (Service Data Unit):** This is like the "item itself" you want to send — the raw data handed down from the upper layer to be processed by the current layer.
- **Header:** When processing data, each layer must add a specific piece of control information to the front of the SDU. It's like slapping on a

"shipping label" to tell the corresponding layer on the receiving end how to handle this data (e.g., packet sequence, encryption method, etc.).

- **PDU (Protocol Data Unit):** Once the item (SDU) is combined with the label (Header) and the packaging for this layer is complete, this entire parcel is called a PDU.

When a packaged PDU from one layer is passed down to the next, it becomes the SDU (the item to be shipped) in the eyes of that next layer. The lower layer then attaches its own header, turning it into a new PDU, and continues passing it down until it becomes an electromagnetic wave emitted at the very bottom. When the base station receives it, the process is a reverse "unpacking": each layer verifies and tears off its respective header, sending the remaining SDU to the upper layer until the most original data is restored.

Earlier, we mentioned that the 5G base station was split into a CU and a DU, but what is the standard for this split? If we lay out this "processing pipeline" shared by both communicating parties and look at the specific tasks handled by these six checkpoints, you will instantly understand why the base station was sliced this way:

- **Radio Resource Control (RRC):** Only the control plane traverses RRC. It is the commander, handling exclusively control signals. When you turn on your phone or disable flight mode, RRC is responsible for the mutual recognition between the phone and the base station. Additionally, RRC manages Handover. Handover means that, imagine you are on a speeding high-speed train, your phone constantly measures the signal strength of nearby base stations and sends the results to the current base station. When the current base station realizes "the base station ahead has a better signal than I do," it issues a command to seamlessly hand your phone over to the next base station.

- **SDAP (Service Data Adaptation Protocol):** This is a brand new protocol born with the 5G network; it did not exist in the 4G era. It only serves the UP, and its task is to map data flows with different QoS requirements coming from the core network (CN) to the corresponding data radio bearers. What does this mean? We mentioned earlier that 5G needs to handle various scenarios, and QoS is like a label explaining what is required. SDAP then throws it onto the corresponding channel. Simply put, it looks at the labels and ensures that "urgent data requiring low latency takes the fast lane, while normal data requiring high bandwidth takes the slow lane," ensuring every type of data receives the transmission treatment that meets its QoS demands.

- **PDCP (Packet Data Convergence Protocol):** This protocol serves both UP and CP, but with different tasks. For UP, it is responsible for "Header Compression (ROHC)." In the network world, a piece of data is preceded by a very long string of "Headers" (like sender and receiver addresses). If you are just sending a "Hi," but the envelope is heavier than the letter itself, it's a huge waste. The ROHC mechanism is a space-saving method that heavily compresses massive headers, allowing electromagnetic waves to carry more truly useful data. For both UP and CP, PDCP performs Ciphering and integrity protection to ensure electromagnetic waves cannot be eavesdropped on or tampered with when intercepted in the air.

- **RLC (Radio Link Control):** On the transmitting end, it is responsible for Segmentation — chopping large packets handed down from upper layers into smaller ones because sometimes the incoming packets are too large for the lower layers to handle. On the receiving end, it is responsible for reassembling the packets and passing them up. At the same time, the RLC layer features an Automatic Repeat Request (ARQ) mechanism. It checks the sequence numbers of arriving packets, and if it finds a packet is missing, it can request the other party to retransmit the data, ensuring transmission reliability. However, this ARQ mechanism is not always turned on, as in some scenarios (like voice calls), a few lost packets don't matter much (at most, it sounds a bit choppy), so this depends on the scenario.

- **MAC (Medium Access Control):** The MAC layer is more like a traffic cop; it decides which phones are allowed to transmit data at any given microsecond. Additionally, the MAC layer contains an error retransmission mechanism faster than RLC's, called Hybrid Automatic Repeat Request (HARQ). We will write another article later to introduce the difference between it and ARQ.
- **PHY (Physical Layer):** If you've studied the course Communication Principles, the vast majority of the content belongs to this layer. This is because this layer involves the most numerous and complex mathematical calculations. Channel Coding and Modulation both occur here, and this is also where signals are converted into electromagnetic waves and transmitted from the antenna.

Having looked at these six core protocol layers (RRC, SDAP, PDCP, RLC, MAC, PHY), we can now go back and solve the earlier puzzle: exactly which part of the base station did 3GPP split into the CU, and which part into the DU?

In the most mainstream 5G base station architecture today, this physical dividing line is drawn precisely between the PDCP layer and the RLC layer:

- **Central Unit (CU):** Responsible for processing the top-level RRC, SDAP, and PDCP. As mentioned earlier, these protocol layers handle control plane command scheduling (RRC), data flow QoS mapping (SDAP), as well as heavy encryption and header compression (PDCP). Although these tasks have complex operational logic, their time sensitivity is relatively low, and they do not require microsecond-level reactions. Therefore, they are highly suitable to be centrally managed in remote equipment rooms or cloud servers.

- **Distributed Unit (DU):** Responsible for processing the bottom-level RLC, MAC, and physical layer (PHY). These layers involve high-frequency packet segmentation and reassembly (RLC), extreme real-time traffic control and dynamic scheduling (MAC), and the lowest-level modulation and coding (PHY). Because these operations must react instantaneously to wireless signals within milliseconds (ms) or even shorter timeframes, they must be deployed at edge sites closest to the antennas (RU).

By precisely slicing the Protocol Stack, the 5G network breaks the monopoly of traditional hardware black boxes, allowing telecom operators to flexibly allocate the computing resources of CU and DU according to different scenario requirements (such as factories needing ultra-low latency, or busy downtown areas needing massive bandwidth) when deploying networks.

In the upcoming 5G series articles, the main focus will revolve around the protocol layers discussed here, explaining some of the fundamental knowledge within the system.

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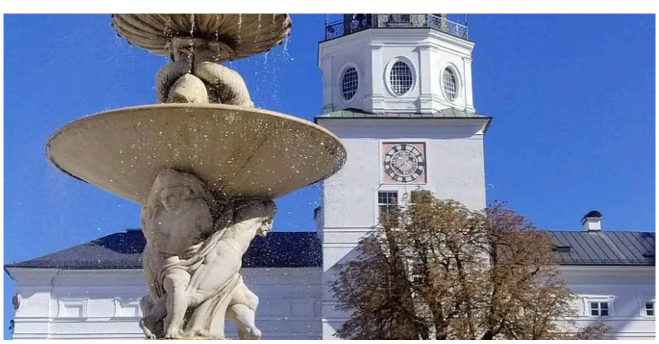

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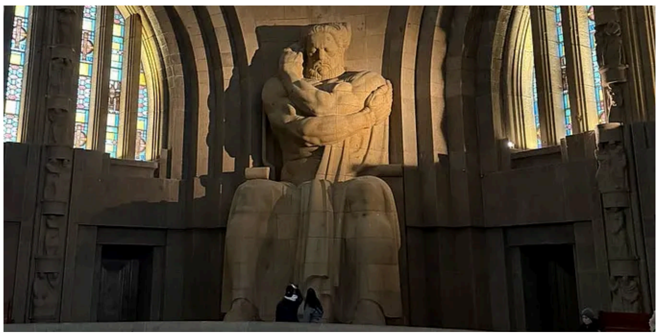

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